

Evaluation of Seguro Popular: Baseline Analysis

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Thanks, in Mexico, to Edith Arano, Carlos Avila, Juan Eugenio Hernández Avila, Mauricio Hernández Avila, Jorge Carreon Manuel Castro, Octavio Gómez Dantes, Sara Odette León, Héctor Hernández Llamas, René Santos Luna, Tania Martínez, Gustavo Olaíz, Maritza Ordaz, Eduardo González Pier, Héctor Peña, Raymundo Pérez, Esteban Puentes, Corina Santangelo, Sergio Sesna, Sara Uriega, Martha María (Mara) Tellez; at Harvard, to Dennis Feehan, Emmanuela Gakidou, Jason Lakin, Diana Lee, Ryan T. Moore, Nirmala Ravishankar, Manett Vargas, and our Panel of Experts, Edmundo Berumen, Luis Felipe Lopez Calva, Nora Claudia Lustig, Thomas Mroz, John Roberto Scott.

The First Results of our Evaluation (Effect of Random Assignment on One Mexican)

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Before Treatment



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(Manett's) Arturo Vargas

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After Treatment



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Evaluation Components

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- National Level Analysis

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- In-depth Focus Groups

Goals of SP & Evaluation Outcome Measures

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Data Sources

- Panel survey ($n = 36,000$) at time 0 and 10 months later

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- Focus group interviews

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 - Places with better doctors and health administration: bigger effects

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 - **Politically Infeasible**: local officials want benefits for their favored areas first

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- If politicians (in a democracy) decide which areas get MAOs
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 - Even if SP has no effect, areas with SP will be healthier

A Feasible Design for Scientific Evaluation

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 - **Methodological**: Drop areas where affiliation had already started

Remaining in study: 148 clusters in 7 states



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 - Same strategy as in most medical intervention studies

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- No randomization at individual level
- Without an evaluation, choices would still be made, but would be arbitrary choices made by local government officials

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 - Consequence: Bias in evaluation conclusions
- We need estimators robust not merely to statistical assumptions but to real world problems

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- Randomization controls for observable and **unobservable confounders**, to a degree
- **Pairing provides failure safeguard**: drop entire pair, and treatment and control groups remain equivalent
- One such failure has already occurred

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 - Repeat until all clusters are used

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- Treatment assignments delivered to state governments

Experimental Design Implementation

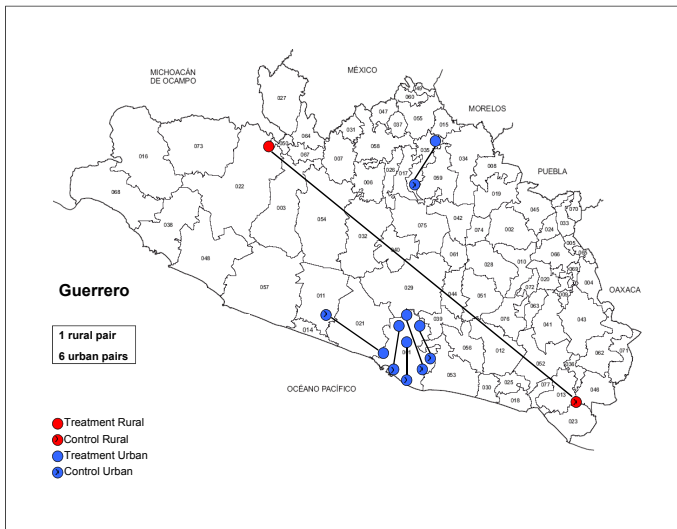
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- Intensive affiliation begins in treatment clusters

Experimental Design Implementation

- At the last moment: Flip coin to choose treatment and control cluster for each pair
- Treatment assignments delivered to state governments
- Intensive affiliation begins in treatment clusters
- 74 matched treatment-control pairs in the evaluation: 55 rural and 19 urban in 7 states

State	Rural Pairs	Urban Pairs	Total
Guerrero	1	6	7
Jalisco	0	1	1
México	35	1	36
Morelos	12	9	21
Oaxaca	3	1	4
San Luis Potosí	2	0	2
Sonora	2	1	3
<i>Total</i>	<i>55</i>	<i>19</i>	<i>74</i>

Matched Pairs, Guerrero

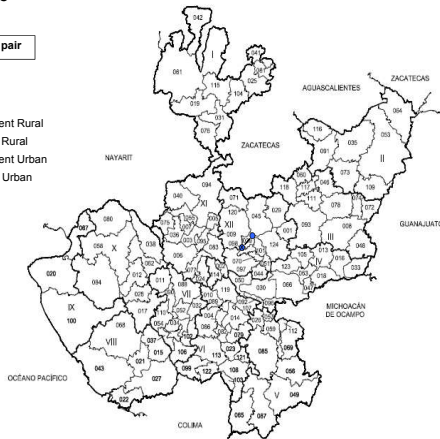


Matched Pairs, Jalisco

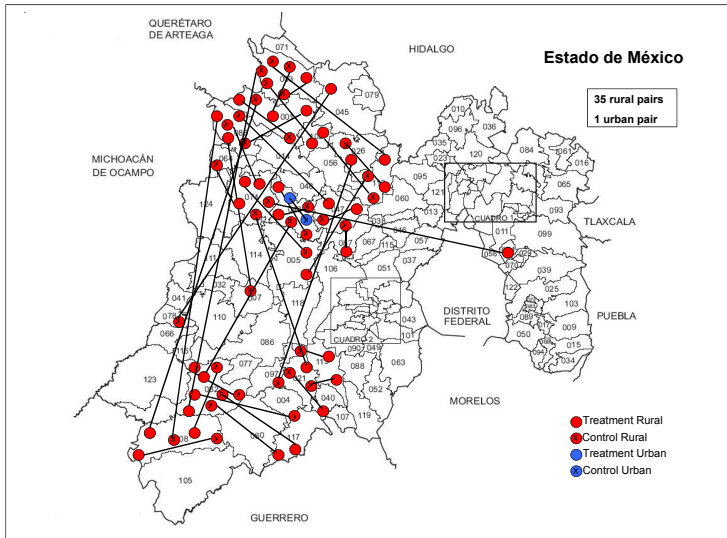
Jalisco

1 urban pair

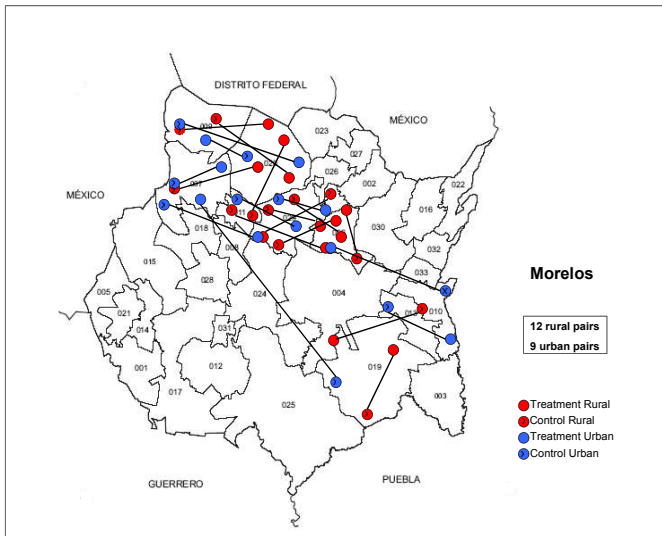
- Treatment Rural
- Control Rural
- Treatment Urban
- Control Urban



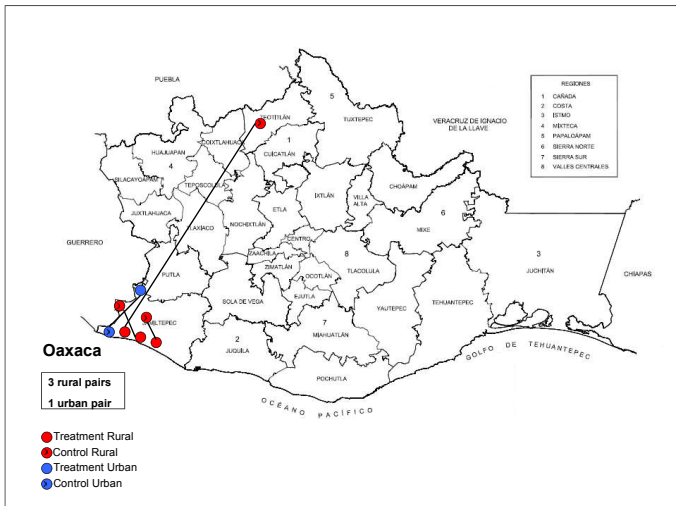
Matched Pairs, Estado de México



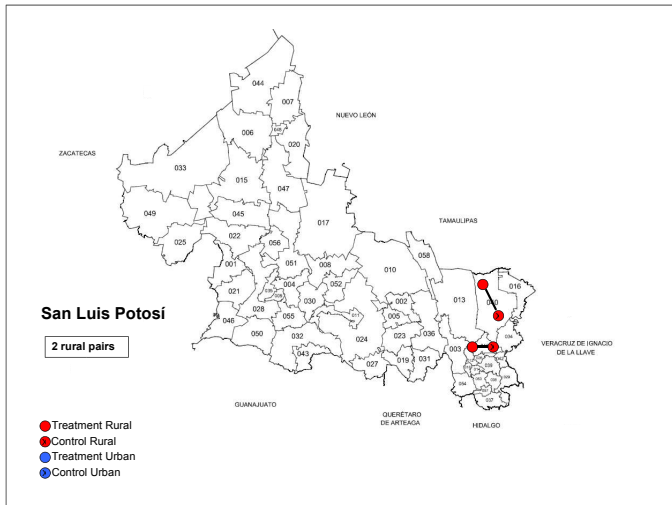
Matched Pairs, Morelos



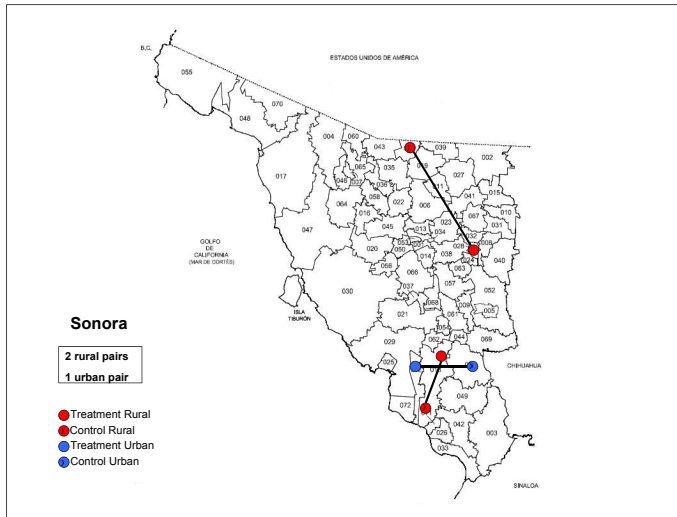
Matched Pairs, Oaxaca



Matched Pairs, San Luis Potosí



Matched Pairs, Sonora



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Design has three parts

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If matching **or** randomization **or** parametric analysis is right, but the other two are wrong, results are still unbiased

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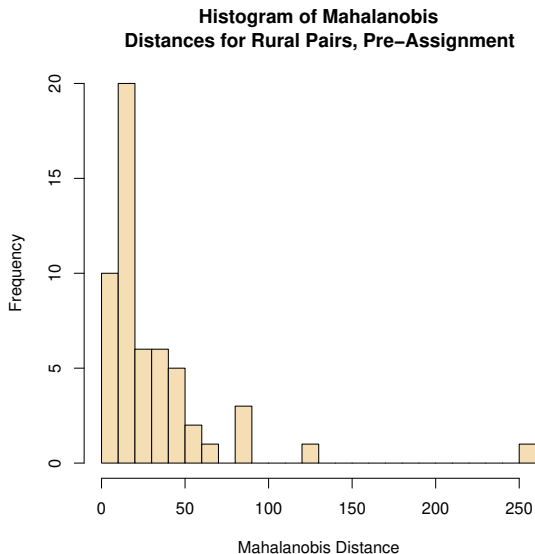
Triple Robustness

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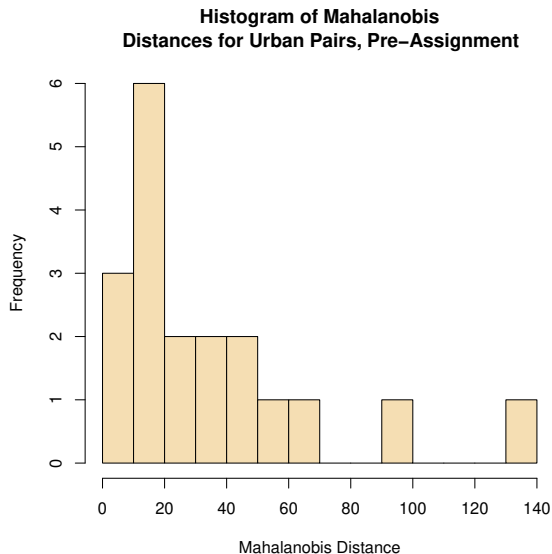
Two Additional Checks if Triple Robustness Fails

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- 2 If we lose pairs, we check for selection bias by rerunning this check

Total Multivariate Distances Within All 55 Rural Pairs

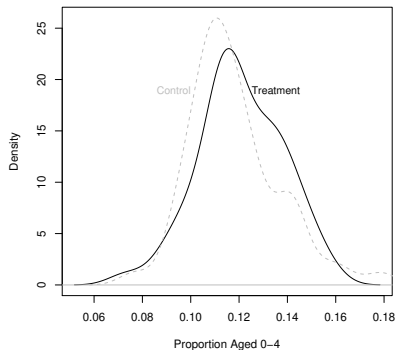


Total Multivariate Distances within All 19 Urban Pairs

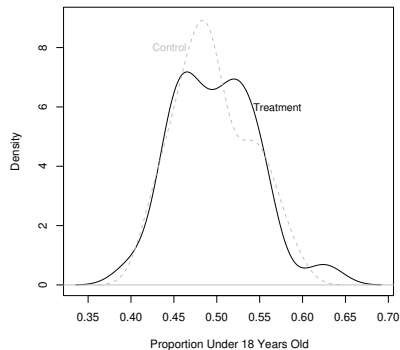


Rural Age Balance After Randomization

Smoothed Histogram of Proportion Aged 0–4, Rural Clusters, Post-Assignment

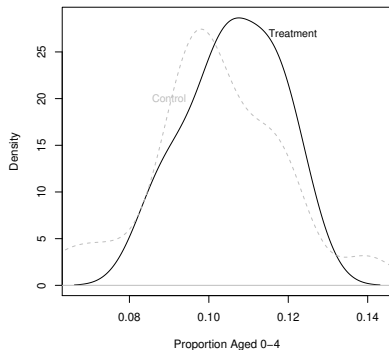


Smoothed Histogram of Proportion Under 18 Years Old, Rural Clusters, Post-Assignment

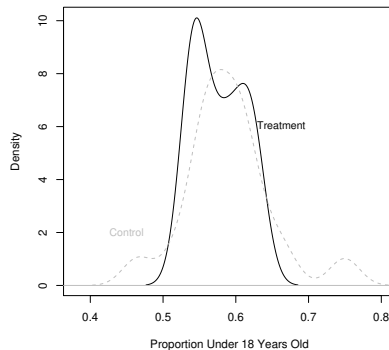


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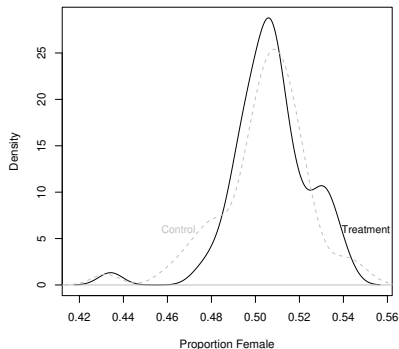


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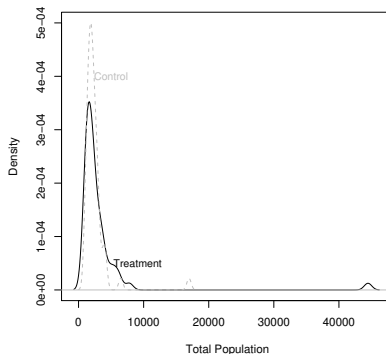


Rural Demographic Balance After Randomization

Smoothed Histogram of Proportion Female, Rural Clusters, Post-Assignment

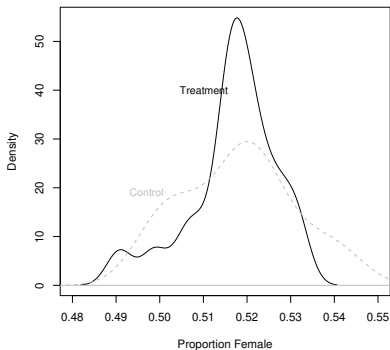


Smoothed Histogram of Total Population, Rural Clusters, Post-Assignment

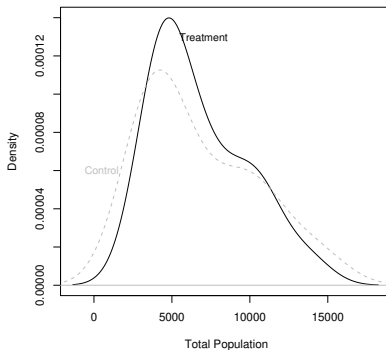


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Smoothed Histogram of Total Population, Urban Clusters, Post-Assignment



Household Survey Design

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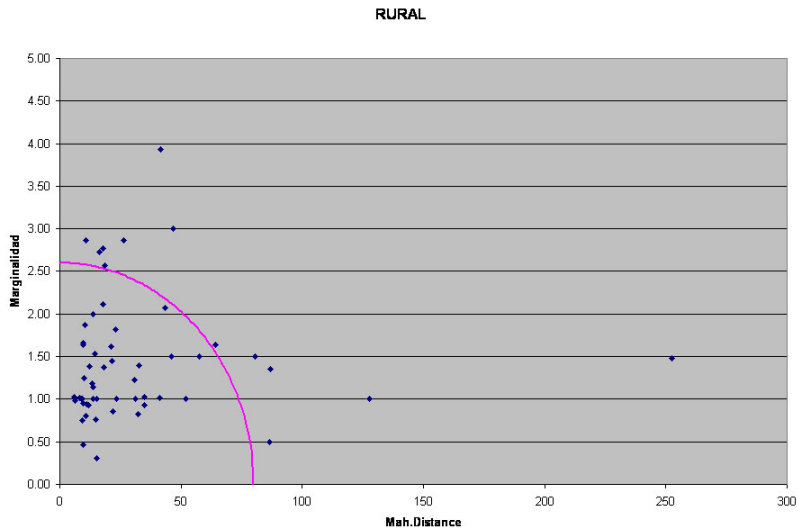
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- Remaining 24 pairs: also used with aggregate outcomes

Choosing Pairs for the Survey



Health Facilities Survey

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Expert Opinions from Ministry Workshop 4/26/06

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(Experts from DGED, INSP, SP, NCGERH, NCESDC)

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Dr. Francisco Garrido

Dr. Esteban Puentes

Dr. Gustavo Olaiz

Laura Mendoza

Lic. Ricardo Forero Paez

Dr. Javier Eduardo Figueroa Zuniga

Dr. Miguel Angel Pena Azuara

Dr. Fernando Escalona Figueroa

Dr. Adolfo Valdez Escobedo

Lic. Luis Carlos Fragoso

Ing. Francisco Salcedo

Dr. Emilio Herrera

Dr. Cuitlahuac Ruiz Matus

Dr. Adrian Delgado Cara

Dra. Liliana Martinez Peafiel

Dra. Ana Mara Sols

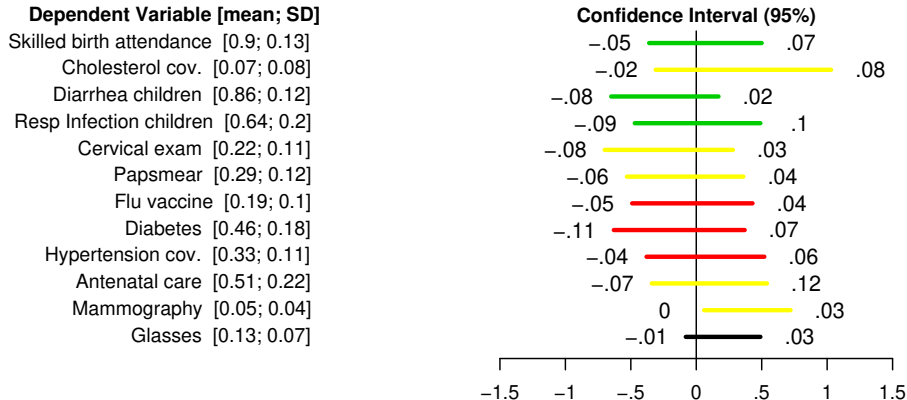
Marisela Cano Bustamante

Dra. Mara Esther Lozano Dvila

Dra. Haidee Caballero Cruz

Effect of SP Rollout at Baseline: 1 of 3

(Expected effects at 10 months: **small**, **medium**, **large**)



Effect of SP Rollout at Baseline: 2 of 3

(Expected effects at 10 months: **small**, **medium**, **large**)

Dependent Variable [mean; SD]

Outpatient visits [1.24; 0.49]

Prescribed drugs [1.2; 0.12]

Waiting time [2.32; 0.23]

SBP [126; 3.05]

Hypertension [0.18; 0.05]

Cholesterol [173; 8.86]

High cholesterol [0.16; 0.09]

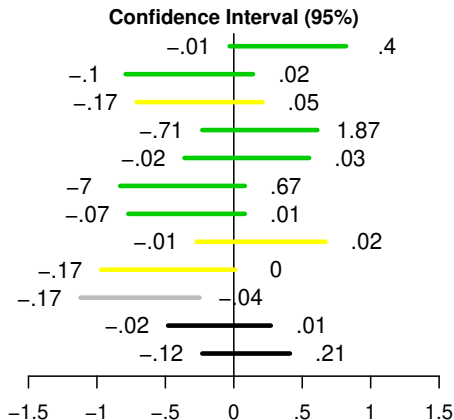
Inpatient visits [0.09; 0.04]

Cleanliness [2.04; 0.17]

Talk privately [2.01; 0.15]

Smoking [0.11; 0.05]

Seatbelt [4.75; 0.5]



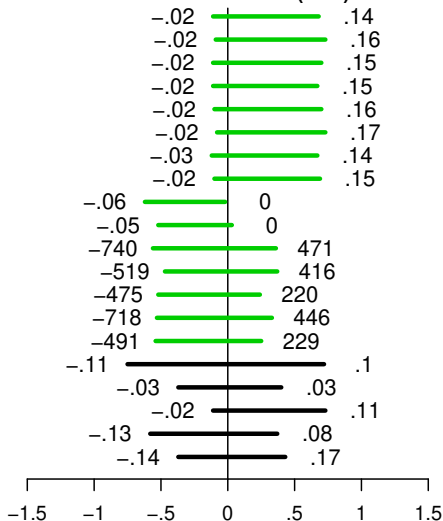
Effect of SP Rollout at Baseline: 3 of 3

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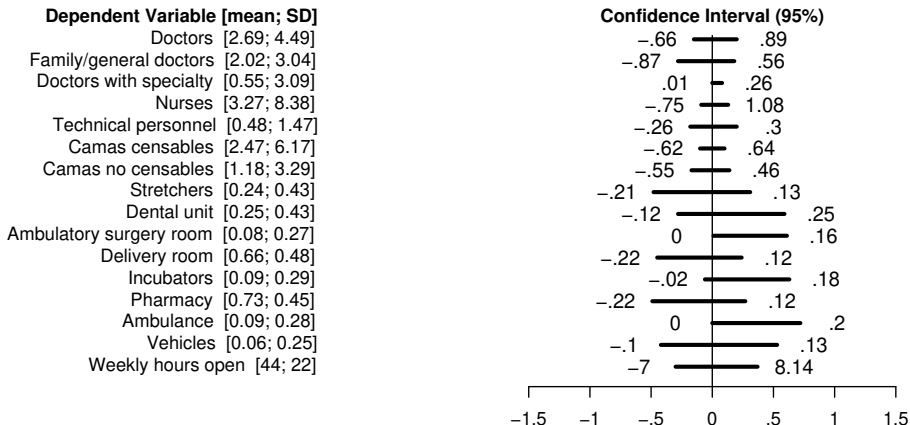
Dependent Variable [mean; SD]

Catastrophic (1,30%)	[0.48; 0.2]
Catastrophic (1,40%)	[0.45; 0.21]
Catastrophic (2,30%)	[0.47; 0.21]
Catastrophic (2,40%)	[0.45; 0.22]
Catastrophic (3,30%)	[0.42; 0.23]
Catastrophic (3,40%)	[0.41; 0.23]
Catastrophic (4,30%)	[0.45; 0.21]
Catastrophic (4,40%)	[0.44; 0.22]
Catastrophic (5,30%)	[0.18; 0.1]
Catastrophic (5,40%)	[0.16; 0.1]
Out of pocket (1)	[3002; 1327]
Out of pocket (2)	[2674; 1113]
Out of pocket (3)	[1488; 915]
Out of pocket (4)	[2320; 1346]
Out of pocket (5)	[1488; 915]
Affiliation	[0.09; 0.14]
Satisfied health	[0.89; 0.08]
Trust local government	[0.29; 0.15]
Reduce rich-poor diff.	[3.42; 0.21]
Privatize electricity	[3.3; 0.39]

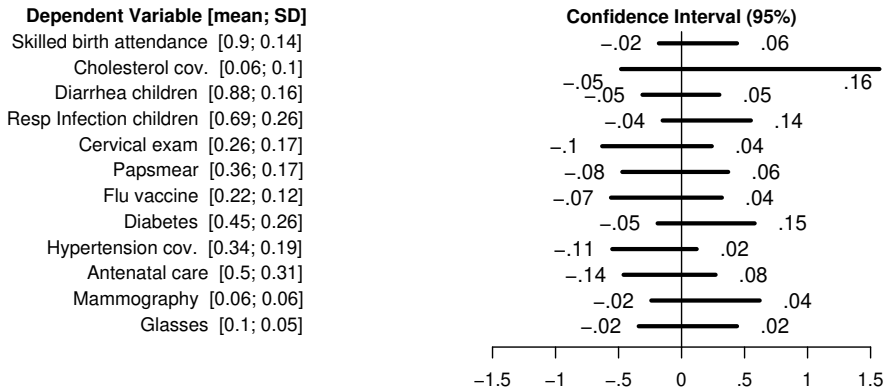
Confidence Interval (95%)



Effect of SP Rollout at Baseline Facilities



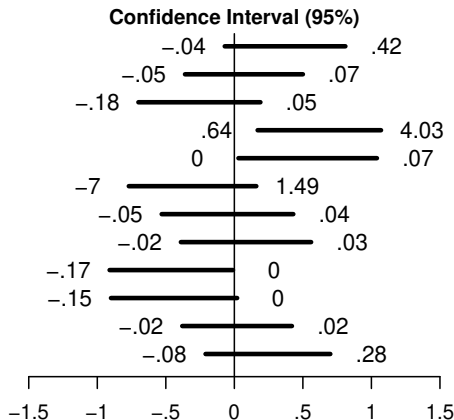
Effect of SP Rollout at Baseline on the Poor: 1 of 3



Effect of SP Rollout at Baseline on the Poor: 2 of 3

Dependent Variable [mean; SD]

Outpatient visits	[1.29; 0.52]
Prescribed drugs	[1.19; 0.14]
Waiting time	[2.31; 0.25]
SBP	[125; 3.78]
Hypertension	[0.17; 0.06]
Cholesterol	[172; 9.12]
High cholesterol	[0.15; 0.09]
Inpatient visits	[0.09; 0.05]
Cleanliness	[2.04; 0.19]
Talk privately	[2.01; 0.17]
Smoking	[0.11; 0.06]
Seatbelt	[4.97; 0.4]

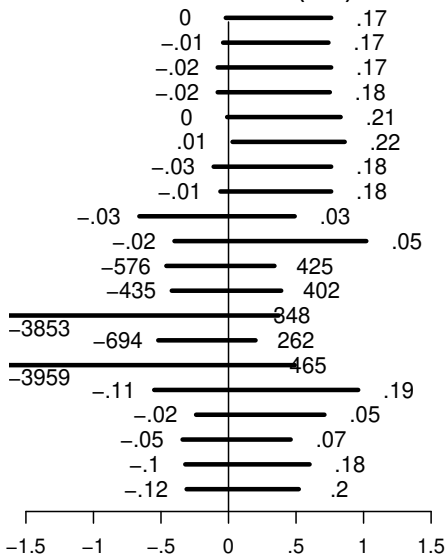


Effect of SP Rollout at Baseline on the Poor: 3 of 3

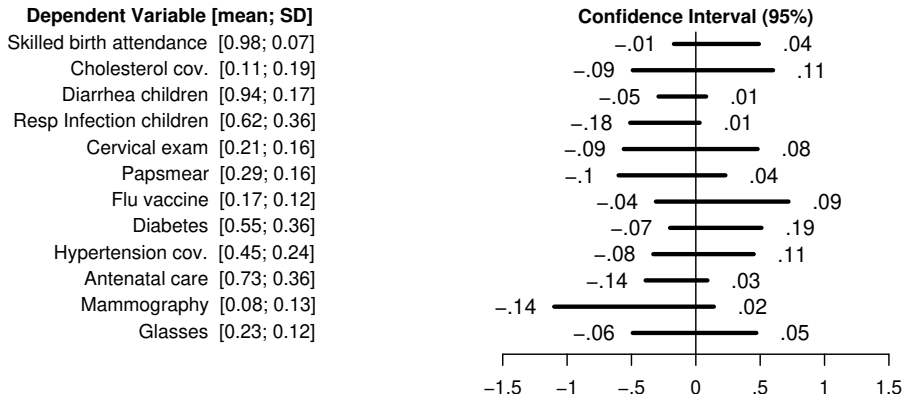
Dependent Variable [mean; SD]

Catastrophic (1,30%) [0.43; 0.22]
 Catastrophic (1,40%) [0.41; 0.23]
 Catastrophic (2,30%) [0.42; 0.22]
 Catastrophic (2,40%) [0.4; 0.23]
 Catastrophic (3,30%) [0.37; 0.25]
 Catastrophic (3,40%) [0.37; 0.25]
 Catastrophic (4,30%) [0.4; 0.23]
 Catastrophic (4,40%) [0.39; 0.24]
 Catastrophic (5,30%) [0.08; 0.05]
 Catastrophic (5,40%) [0.06; 0.05]
 Out of pocket (1) [2552; 1263]
 Out of pocket (2) [2308; 1035]
 Out of pocket (3) [1316; 951]
 Out of pocket (4) [1931; 1330]
 Out of pocket (5) [1316; 951]
 Affiliation [0.13; 0.2]
 Satisfied health [0.9; 0.07]
 Trust local government [0.3; 0.16]
 Reduce rich-poor diff. [3.39; 0.3]
 Privatize electricity [3.23; 0.39]

Confidence Interval (95%)



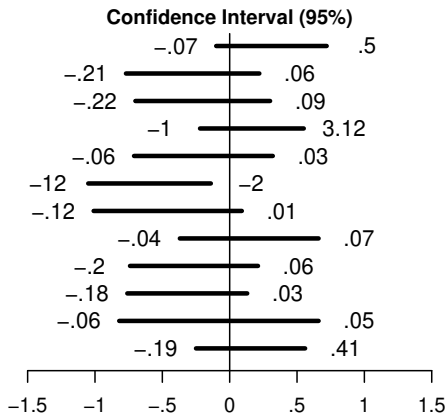
Effect of SP Rollout at Baseline on the Wealthy: 1 of 3



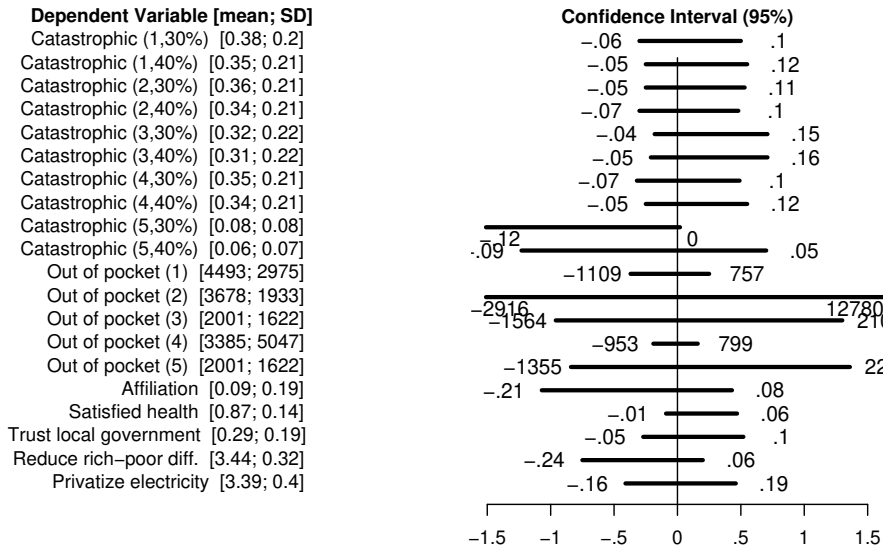
Effect of SP Rollout at Baseline on the Wealthy: 2 of 3

Dependent Variable [mean; SD]

Outpatient visits [1.45; 0.7]
Prescribed drugs [1.18; 0.28]
Waiting time [2.31; 0.31]
SBP [125; 5.69]
Hypertension [0.16; 0.09]
Cholesterol [175; 11]
High cholesterol [0.18; 0.12]
Inpatient visits [0.11; 0.1]
Cleanliness [1.99; 0.27]
Talk privately [2; 0.24]
Smoking [0.11; 0.08]
Seatbelt [4; 0.74]

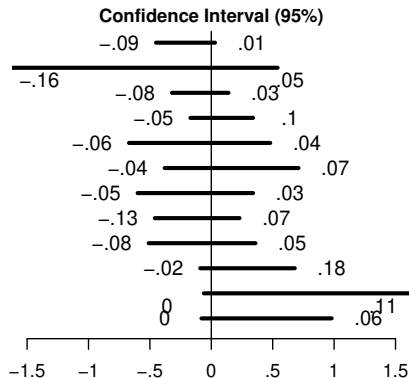


Effect of SP Rollout at Baseline on the Wealthy: 3 of 3



Effect of SP Rollout at Baseline on Others: 1 of 3

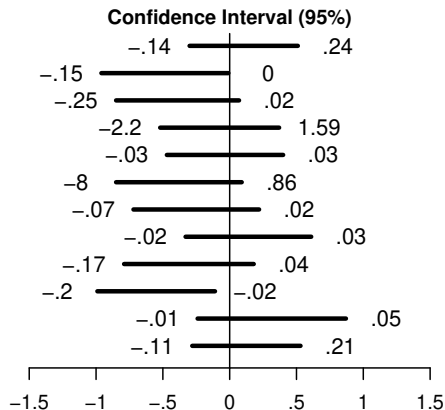
Dependent Variable [mean; SD]
Skilled birth attendance [0.89; 0.2]
Cholesterol cov. [0.06; 0.08]
Diarrhea children [0.8; 0.24]
Resp Infection children [0.63; 0.3]
Cervical exam [0.16; 0.09]
Papsmear [0.21; 0.1]
Flu vaccine [0.14; 0.09]
Diabetes [0.39; 0.28]
Hypertension cov. [0.27; 0.15]
Antenatal care [0.53; 0.27]
Mammography [0.04; 0.04]
Glasses [0.11; 0.06]



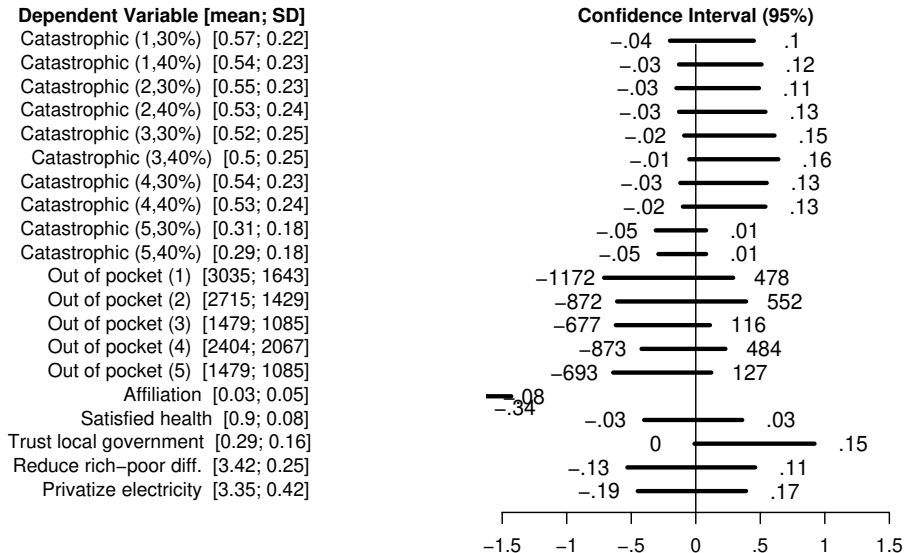
Effect of SP Rollout at Baseline on Others: 2 of 3

Dependent Variable [mean; SD]

Outpatient visits [1.08; 0.46]
Prescribed drugs [1.22; 0.16]
Waiting time [2.31; 0.29]
SBP [126; 4.26]
Hypertension [0.19; 0.07]
Cholesterol [173; 9.58]
High cholesterol [0.16; 0.1]
Inpatient visits [0.1; 0.05]
Cleanliness [2.04; 0.21]
Talk privately [2.02; 0.21]
Smoking [0.11; 0.06]
Seatbelt [4.86; 0.4]



Effect of SP Rollout at Baseline on Others: 3 of 3



For more information

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